

Enhancing black-box membership inference attacks in federated learning

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ABSTRACT

With the widespread deployment of machine learning models in privacy-sensitive domains such as healthcare and finance, the risk of training data leakage has attracted increasing attention. As a fundamental approach for evaluating model privacy leakage, Membership Inference Attack (MIA) has been extensively studied in distributed learning scenarios such as Federated Learning (FL). However, under black-box settings, attackers face severe challenges, including the unavailability of real non-member samples and the inaccessibility of target model architectures, which limit the generalization and accuracy of existing methods. To address these limitations, this paper proposes a DCGAN-enhanced black-box MIA framework, whose innovations are reflected in three major aspects: (1) a discriminator-guided pseudo-sample filtering mechanism that ensures the authenticity and diversity of non-member data; (2) a multi-shadow-model softmax high-dimensional concatenation strategy, which fuses the softmax probability outputs from multiple shadow models to construct discriminative high-dimensional attack representations; and (3) a SMOTE-based balancing module designed to mitigate class imbalance and further improve the generalization of the attack model. The proposed framework significantly enhances the discriminative capability and robustness of black-box MIAs without accessing the internal parameters or training procedures of the target model. Extensive experiments demonstrate that our method consistently outperforms state-of-the-art baselines across multiple federated learning protocols (FedAvg, FedMD, and FedProx) and benchmark datasets (CIFAR-10, CIFAR-100, Fashion-MNIST, and SVHN), achieving an accuracy of 0.9897, an AUC of 0.9899, and a TPR@FPR=1% of 0.9967. These results verify the robustness, generalizability, and wide applicability of the proposed framework, providing a systematic and scalable solution for privacy evaluation in federated learning environments.

1. Introduction

With the increasing stringency of data privacy regulations, *Federated Learning* (FL) [1] has been widely adopted in privacy-sensitive domains such as healthcare and finance due to its decentralized training paradigm where “data never leaves the local device” [2,3]. Although FL avoids centralized storage and direct transmission of raw data — seemingly offering inherent privacy protection — extensive studies have shown that the shared global model parameters and prediction outputs can still inadvertently leak sensitive information from local training data [4]. Particularly under black-box or unsupervised settings, adversaries can craft targeted inference strategies solely based on model outputs to determine whether a specific sample has participated in the training process, thus compromising individual data privacy. *Membership Inference Attacks* (MIA) [5] represent a typical form of such privacy risks. These attacks aim to infer whether a given data instance was part of the training set, potentially revealing sensitive membership status or source information. In FL scenarios, MIA can bypass traditional access

control and derive privacy leakage purely from model behavior, making it a key approach for assessing the privacy vulnerability of federated models [6].

Existing MIA methods can be broadly categorized into white-box, gray-box, and black-box attacks [7]. White-box attacks assume complete access to model parameters, gradients, or training procedures, allowing attackers to analyze internal representations and achieve high attack accuracy. Gray-box attacks allow partial access to model architecture, training settings, or auxiliary data, enabling attackers to reconstruct shadow models or leverage prior knowledge. However, in realistic federated learning [1] settings, attackers often lack access to the model internals and can only observe output predictions, leading to a stricter yet more practical *black-box* attack setting.

In recent years, research on black-box MIA has progressed rapidly, with various strategies proposed to enhance attack effectiveness under realistic federated constraints. Representative approaches can be grouped into three main directions: statistical-feature-based inference,

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trigger-based manipulation, and generative-model-based augmentation. The first two rely on extended model interaction or embedded mechanisms, which limit practicality and violate strict black-box assumptions. Generative models, such as GANs [8], alleviate the scarcity of non-member samples by synthesizing pseudo data, but without robust quality control, these samples may become overly realistic or distributionally misaligned, introducing label noise and degrading generalization. Moreover, existing studies often neglect the class imbalance between member and non-member samples, further reducing attack stability and realism in resource-constrained environments.

To address these challenges, we propose an enhanced black-box MIA framework that integrates pseudo non-member sample filtering and multi-shadow-model output fusion to balance attack effectiveness and feasibility under black-box constraints. The main contributions of this paper are as follows:

- We propose a discriminator-guided quality control mechanism that leverages DCGAN [9] to synthesize and filter pseudo non-member samples [10]. This approach mitigates the scarcity of real negative data while avoiding overly realistic samples that may introduce label noise, thereby enhancing both training stability and label consistency.
- We propose a high-dimensional statistical feature construction mechanism that aggregates outputs from multiple shadow models [11]. Seven statistical features are extracted from the softmax predictions and concatenated across models to preserve inter-model prediction heterogeneity and enhance feature discriminability. The SMOTE oversampling technique [12] is employed to alleviate class imbalance. An XGBoost-based [13] attack model is trained with nonlinear boosting and ensemble learning strategies, effectively fitting complex decision boundaries. Finally, we evaluate generalization performance and feature sensitivity to demonstrate the stability and practical utility of the proposed method.

2. Related work

Although federated learning (FL) avoids centralized data transmission, it remains vulnerable to privacy leakage through model outputs. Under the black-box setting, membership inference attacks (MIA) can determine whether a sample was used in training by analyzing model predictions, without requiring access to training data or model parameters. As such, MIA has emerged as a core privacy threat in federated learning.

To address the difficulty of obtaining non-member samples in black-box FL settings, researchers have proposed using generative models (e.g., GANs, VAEs) to synthesize pseudo non-member samples, alleviating the scarcity of negative samples in attacker training. Hilprecht et al. [14] introduced a membership inference method based on Monte Carlo sampling and reconstruction distance, employing GANs to generate auxiliary samples, which significantly improved attack performance under data-limited conditions. However, they lacked control over the distribution and label purity of the generated samples, often leading to ambiguous decision boundaries and degraded generalization. Yan et al. [15] further introduced a VAE-GAN framework to enhance data diversity, but their experiments also revealed that the absence of quality filtering could misguide the attack model and impair performance. In addition, the 2023 study titled *Synthetic is All You Need* [16] demonstrated that MIA can be successfully conducted solely with synthetic data, highlighting the potential of generative models in attack construction. A 2024 survey titled *A Systematic Review of Federated Generative Models* [17] systematically reviewed the application of generative models in FL, pointing out that while GANs, VAEs [18], and diffusion models [19] have been widely applied, most approaches lack effective quality control mechanisms, resulting in distribution mismatches and limited attack interpretability. In summary, although generative

data augmentation has made progress in mitigating negative sample scarcity, constructing high-quality, distribution-controllable pseudo non-member samples remains a critical challenge in black-box MIA.

Beyond academic datasets, federated learning has also been applied to industrial and IoT-based scenarios. Ali et al. [20] presented a CNN-enhanced federated learning framework for industrial IoT intrusion detection, achieving both privacy preservation and high detection performance while significantly reducing communication costs. Similarly, Ali and Islam [21] proposed a sustainable forest fire detection framework within the Green IoT paradigm, integrating federated learning with DenseNet-based feature extraction to enable energy-efficient and privacy-preserving environmental monitoring. These practical studies demonstrate the applicability and scalability of federated systems under adversarial and real-world conditions, providing valuable context for exploring privacy vulnerabilities and robustness issues in federated architectures.

In black-box federated attack scenarios, adversaries face stricter constraints: no access to local data, intermediate parameters, or model structures—only limited rounds of global model predictions are observable. These limitations greatly increase the difficulty of conducting MIA. In early centralized scenarios, Shokri et al. [22] proposed the first MIA framework based on multiple shadow models to simulate the target model's behavior, training an attack model to distinguish member from non-member samples, thereby establishing a foundational paradigm. However, this method assumes white-box access and full training data, making it unsuitable for pure black-box FL settings. To overcome this, various methods have been proposed for black-box FL attacks. Hu et al. [23] introduced the MIB method, embedding backdoor triggers (e.g., fixed pixel blocks) during training to induce distinct predictions for member samples. While effective in controlled environments, MIB violates the black-box assumption and poses deployment risks. Chen et al. [24] employed data poisoning combined with confidence modeling to manipulate prediction confidence for member detection, but its reliance on input interference limits its practicality. He et al. [25] proposed a method based on confidence time series, extracting softmax sequence features (e.g., mean, slope, variance) across training rounds and applying an AdaBoost classifier for inference. However, the need for continual model access restricts long-term applicability. Nguyen et al. [26] proposed a clustering-based membership inference algorithm named *DocMIA*, which analyzes the distributional differences between members and non-members in high-dimensional feature space and leverages unsupervised clustering to infer membership status. This method is designed for DocVQA models and operates effectively under an auxiliary-data-free setting. Although *DocMIA* achieves promising results on document-level multimodal tasks, it has certain limitations, including a strong dependence on the quality and quantity of question-answer pairs, high computational overhead during inference, and limited generalizability to non-multimodal models or standard classification tasks.

In contrast, our proposed feature fusion strategy achieves better generality and robustness under strict black-box constraints. Specifically, without relying on backdoor injection, data poisoning, or multi-round model access, we construct joint response vectors by fusing softmax outputs from multiple shadow models, significantly enhancing the discriminability and representational power of the attack features. This strategy not only expands the dimensionality of attack signals within black-box limits but also improves generalization across heterogeneous model outputs, offering a more feasible and stable attack path in federated MIA.

To address the above challenges, we propose a black-box MIA enhancement framework based on pseudo non-member sample filtering and multi-shadow-model output fusion. Our method includes: (1) generating pseudo non-member samples via DCGAN and filtering overly "realistic" samples using a discriminator scoring mechanism to ensure label purity; (2) fusing softmax outputs from multiple shadow models to

construct high-dimensional statistical features for expressive attack representation; and (3) applying SMOTE to mitigate class imbalance and training a robust XGBoost-based attack classifier. This integrated framework jointly improves attack performance and generalization through coordinated enhancement in sample generation, feature modeling, and attack optimization.

3. Three-stage black-box membership inference attack framework

3.1. Overview of the technical framework

We propose a comprehensive black-box MIA framework for federated learning, consisting of pseudo non-member construction, multi-shadow model fusion, feature extraction, and attack model training. As illustrated in Fig. 1, the overall process is divided into three sequential stages: (1) pseudo non-member sample generation and filtering, (2) multi-shadow model fusion and feature extraction, and (3) attack model construction and optimization.

In the first stage, to address the scarcity of non-member samples, we employ DCGAN to synthesize pseudo non-member samples based on an auxiliary dataset. A discriminator is used to score each generated sample, and a threshold (e.g., 0.6) is applied to filter out overly realistic samples. This scoring mechanism helps ensure the label purity of pseudo non-members and enhances the representativeness and discriminative power of the negative class samples.

During the second stage, we construct a multi-shadow model architecture where each model is trained on an independently partitioned dataset to simulate the target model's prediction behavior from different perspectives. In the inference phase, all shadow models produce softmax predictions for each sample. The resulting probability vectors are concatenated in order, forming a higher-dimensional composite prediction vector. This concatenation preserves the diverse prediction patterns across models and highlights subtle differences in their output behaviors, thereby enhancing the expressiveness and separability of the attack features. On this high-dimensional concatenated output, we further extract seven key statistical attack features, including maximum confidence, confidence gap, entropy, and standard deviation, to capture the behavioral discrepancies between member and non-member samples in the output space. These features serve as fine-grained and discriminative inputs for the subsequent attacker model.

In the third stage, to alleviate class imbalance, we introduce the SMOTE technique to enrich the diversity of the positive (member) class. We then employ XGBoost as the attack classifier for model training and inference. This combination not only improves attack performance but also enhances the model's generalization capability.

Overall, the proposed framework integrates data generation, feature fusion, and model optimization into a unified three-stage pipeline. The first stage focuses on pseudo non-member generation and filtering using DCGAN and discriminator scoring; the second stage enhances feature expressiveness through multi-shadow-model fusion; and the third stage strengthens classification robustness via SMOTE-based balancing and XGBoost optimization. Together, these stages progressively refine the attack process from data synthesis to model inference.

3.2. Pseudo non-member sample generation and filtering mechanism

In black-box membership inference attacks within federated learning, non-member samples are often difficult to obtain directly, resulting in a severe class imbalance between member and non-member samples during the training of the attack model. This imbalance negatively impacts both the performance and generalization ability of the attacker. To address this issue, we propose a pseudo non-member sample generation and filtering mechanism based on Deep Convolutional Generative Adversarial Networks (DCGAN). DCGAN employs an adversarial learning paradigm between a generator and a discriminator, enabling the generator to learn the underlying data manifold and synthesize diverse,

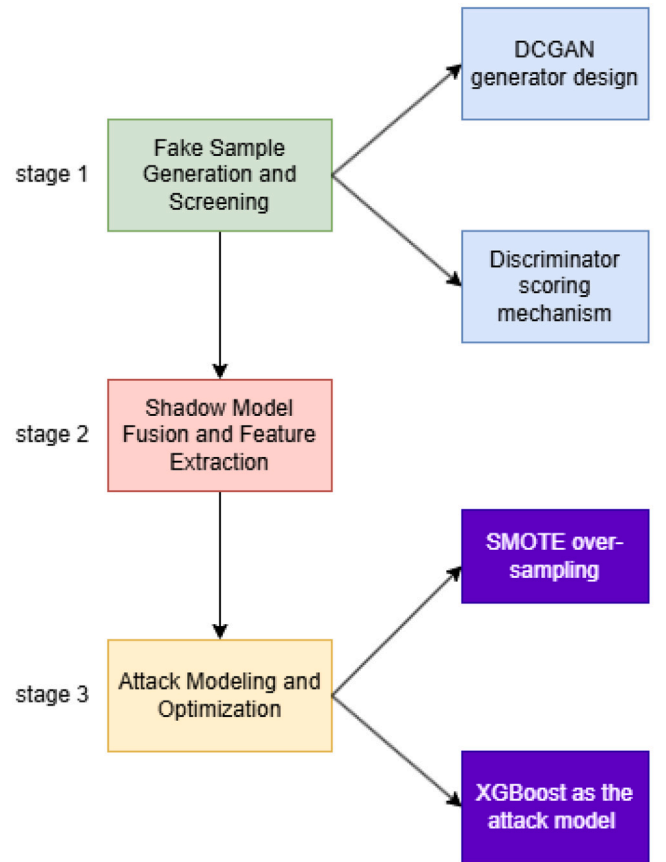


Fig. 1. Three-stage framework of the proposed black-box MIA in federated learning. DCGAN generates and filters pseudo non-members, shadow models provide fused features, and XGBoost with SMOTE forms the final attack model.

high-quality pseudo samples that approximate the real data distribution. Compared with traditional random perturbation or oversampling methods, DCGAN captures richer spatial and semantic information, which helps generate more realistic yet distributionally distinct non-member samples. The goal is to synthesize high-quality negative samples from auxiliary data and apply a discriminator-based quality control strategy to filter pseudo non-member samples. This approach enhances the label purity and discriminative power of the training set, thereby improving the overall effectiveness of the attack.

3.2.1. Pseudo non-member sample generation mechanism

Fig. 2 illustrates the auxiliary data construction process for membership inference attacks based on DCGAN. First, real non-member samples are selected from an auxiliary dataset and used as input to the generator. The generator learns the target distribution and outputs pseudo non-member samples $G(z)$. These generated samples are used in two ways: (1) they are fed into the discriminator and compared with real samples for adversarial training, and (2) they are used to compute the adversarial loss $\mathcal{L}_G$ to update the generator's parameters.

To control the quality and label purity of pseudo non-member samples, a discriminator-based scoring mechanism is introduced. Only pseudo samples with discriminator scores below a threshold t , i.e., $D(G(x)) < t$, are retained, as they are considered to have not excessively approximated real samples, thereby reducing label contamination. The retained pseudo non-member samples are then concatenated with the original real non-member samples to form an enhanced negative sample set for training the attack model.

However, the generated samples may vary in quality, requiring a scoring-based filtering mechanism to ensure label purity, which is introduced in the next subsection.

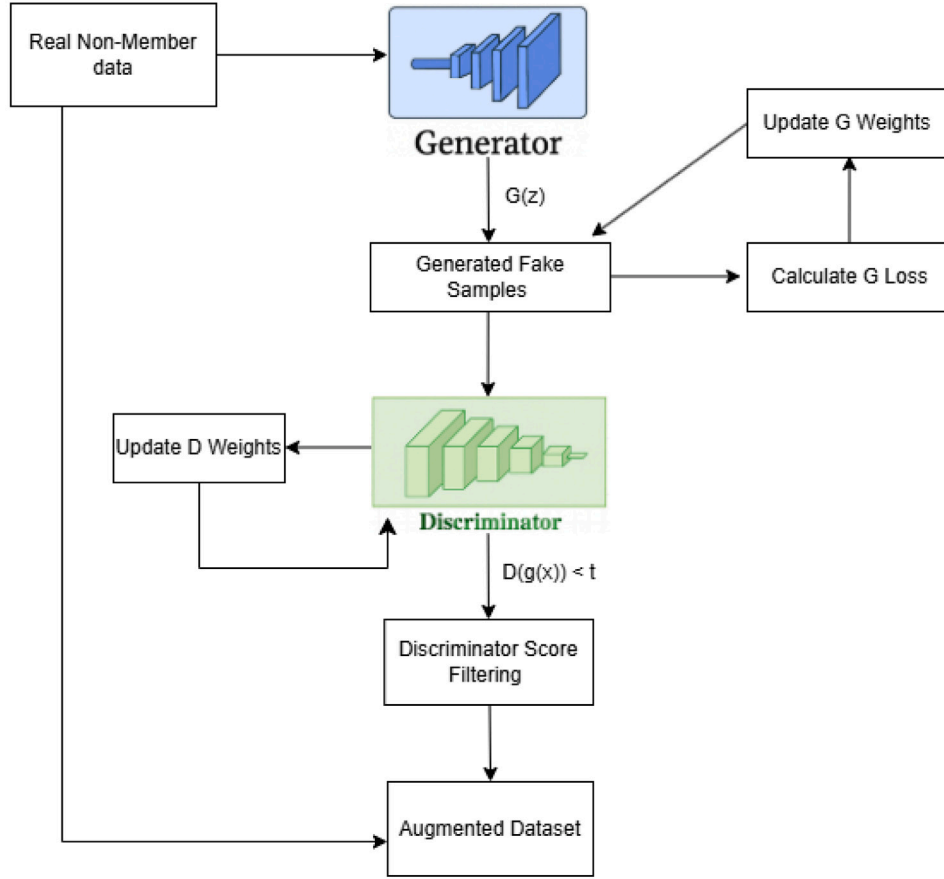


Fig. 2. DCGAN-based pseudo non-member generation and filtering process. Real non-member data train the generator to produce pseudo samples $G(z)$. The discriminator scores $D(G(x))$ and retains those below threshold t to form an augmented negative dataset.

3.2.2. Discriminator scoring and pseudo non-member sample filtering mechanism

In black-box membership inference attacks under federated learning, the performance of the attack model heavily depends on the quality and representativeness of negative samples (i.e., non-member samples). Since real non-member samples are inaccessible, traditional methods often rely on unscreened generated samples or test set data as negatives, which tends to blur the boundary between members and non-members, ultimately hindering the learning of a well-defined decision boundary for the attacker.

To address this issue, we propose a discriminator-driven pseudo non-member sample filtering mechanism. This mechanism utilizes numerical scoring and distribution control to evaluate the quality of samples generated by DCGAN. By selectively incorporating only qualified samples, the approach significantly enhances the representativeness and label purity of the negative sample set.

Discriminator score as an authenticity indicator for pseudo non-member samples. After training the Generative Adversarial Network (GAN), the discriminator $D : \mathcal{X} \rightarrow [0, 1]$ learns a representation space for distinguishing real samples from generated pseudo non-member samples. The output $D(x)$ can be interpreted as a *confidence score* indicating the likelihood that a sample belongs to the true data distribution, defined as:

$$s_i = D(\tilde{x}_i), \quad s_i \in [0, 1]. \quad (1)$$

Here, $\tilde{x}_i = G(z_i)$ denotes a pseudo non-member sample generated by the generator $G : \mathbb{R}^d \rightarrow \mathcal{X} \subseteq \mathbb{R}^{H \times W \times C}$ from a latent variable $z_i \sim \mathcal{N}(0, I_d)$. A higher score s_i indicates that the discriminator considers the sample more “real”, meaning its distribution is closer to the training

data. However, in membership inference attacks, the attacker requires negative samples that are distinguishable from member samples. If pseudo non-member samples resemble real data too closely, their discriminative utility as negatives is diminished. Compared to traditional approaches that directly use generated samples or randomly combine them, the proposed discriminator-guided filtering mechanism significantly enhances the separability between member and non-member samples in the data distribution space. It mitigates the issue of label noise, improves the structural integrity of the attack training set, and stabilizes the learning process of the attack model. This mechanism constitutes one of the key innovations of our proposed framework, providing a higher-quality foundation for subsequent attack feature modeling and discriminator training.

Threshold-based sample filtering strategy. To mitigate the above issue, we introduce a filtering threshold $\tau \in [0, 1]$ to constrain the upper bound of pseudo non-member sample authenticity, where τ denotes the discriminator score threshold. Only generated samples with scores below τ are retained as negative inputs:

$$\mathcal{X}_{\text{GAN}} = \{\tilde{x}_i \mid D(\tilde{x}_i) \leq \tau\}. \quad (2)$$

This filtering process selects a subset of generated samples that are sufficiently distant from the training data in the distributional space, thereby reinforcing the boundary positioning of negative samples. In experiments, we empirically set the threshold to $\tau = 0.6$, based on the observed bimodal distribution of discriminator outputs. The histogram of $D(\tilde{x}_i)$ shows a clear inflection point near 0.6, marking the transition between overly realistic and synthetically distinct regions. This data-driven criterion effectively filters out pseudo non-member samples that are overly similar to the real data distribution while

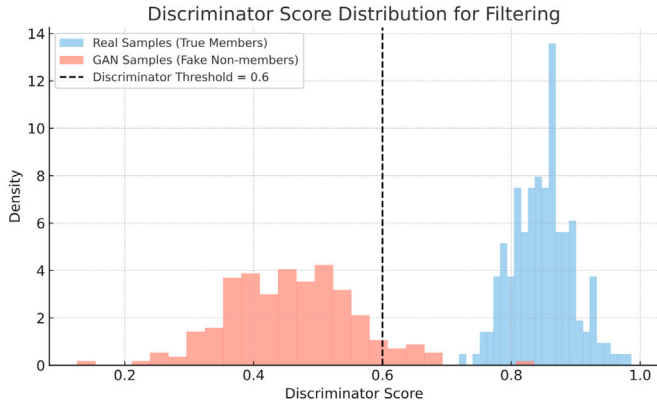


Fig. 3. Discriminator score distribution for real and fake samples. Real members have higher scores, while GAN-generated non-members show lower ones. A threshold τ filters overly realistic samples to improve non-member discriminability.

preserving diversity and generalizability, providing a transparent and reproducible filtering mechanism consistent with recent studies on transparent decision-making in federated learning [27].

Distribution visualization and decision boundary refinement. Fig. 3 illustrates the distribution of discriminator scores for real and generated samples. As shown, real samples cluster in the high-score region, while low-score samples tend to deviate further from the training data distribution. By introducing τ as a decision boundary, we can effectively avoid the risk of “label contamination” from high-confidence generated samples. Simultaneously, the retained negatives form a clearer boundary in probability space, which facilitates the attack model in learning a more precise discrimination strategy.

Contribution of the filtering mechanism to attack training set construction. Finally, the filtered pseudo non-member samples $\mathcal{X}_{\text{GAN}}$, together with the test set samples $\mathcal{X}_{\text{test}}$, jointly constitute the negative class of the attack training dataset:

$$\mathcal{D}_{\text{non}} = \{(\tilde{x}_i, y_i = 0) \mid \tilde{x}_i \in \mathcal{X}_{\text{GAN}}\} \cup \{(x_j^{\text{test}}, y_j = 0)\} \quad (3)$$

This is then merged with the member sample set $\mathcal{D}_{\text{member}} = \{(x_k, y_k = 1)\}$ to form the complete attack training set:

$$\mathcal{D}_{\text{attack}} = \mathcal{D}_{\text{member}} \cup \mathcal{D}_{\text{non}} \quad (4)$$

Compared to traditional approaches that directly use generated samples or randomly combine them, the proposed discriminator-guided filtering mechanism significantly enhances the separability between member and non-member samples in the data distribution space. It mitigates the issue of label noise, improves the structural integrity of the attack training set, and stabilizes the learning process of the attack model. This mechanism constitutes one of the key innovations of our proposed framework, providing a higher-quality foundation for subsequent attack feature modeling and discriminator training.

3.2.3. Mechanism for constructing the non-member training set via filtered pseudo non-member samples and auxiliary data

The generation and filtering of pseudo non-member samples represent only one critical step in the construction of the attack training dataset. To complete the dataset structure, it is also necessary to incorporate real non-member samples. In practical black-box federated attack scenarios, the adversary can typically access a portion of the test data that was not involved in training, which serves as an additional source of non-member samples.

To further enhance the diversity and distributional completeness of the negative class, this study integrates the discriminator-filtered

pseudo non-member samples with the auxiliary test set samples, thereby constructing a high-confidence and structurally balanced non-member training dataset.

Construction logic of auxiliary non-member samples. Let the set of pseudo non-member samples filtered by the discriminator be defined as:

$$\mathcal{X}_{\text{GAN}} = \{\tilde{x}_i \mid D(\tilde{x}_i) \leq \tau\} \quad (5)$$

Meanwhile, denote the real non-member sample set from the test dataset as $\mathcal{X}_{\text{test}}$. The complete negative training dataset is then constructed as:

$$\mathcal{D}_{\text{non}} = \{(\tilde{x}_i, y_i = 0) \mid \tilde{x}_i \in \mathcal{X}_{\text{GAN}}\} \cup \{(x_j, y_j = 0) \mid x_j \in \mathcal{X}_{\text{test}}\} \quad (6)$$

This hybrid mechanism combines the discriminative power of generated samples with the authenticity of real non-member samples, avoiding distributional bias caused by relying solely on generated data.

Label purity and decision boundary control strategy. By setting a strict discriminator threshold τ to filter generated samples, we effectively cap their “realism”, ensuring they remain outside the member distribution and reducing the risk of negative label contamination. Meanwhile, test samples in the auxiliary set naturally did not participate in model training and thus inherently possess non-member label attributes. The combination of both sources results in a negative sample distribution that is semantically closer to the generalized concept of “non-members” while exhibiting greater diversity and separability in statistical space.

Final attack training dataset construction. Combining the member sample set $\mathcal{D}_{\text{member}} = \{(x_k, y_k = 1)\}$, the final attack training dataset is formulated as:

$$\mathcal{D}_{\text{attack}} = \mathcal{D}_{\text{member}} \cup \mathcal{D}_{\text{non}} \quad (7)$$

This dataset incorporates three levels of design optimization:

- Mitigates the scarcity of non-member samples using DCGAN-generated data;
- Improves negative label credibility through discriminator-based filtering;
- Enhances diversity and naturalness of the negative class by incorporating auxiliary test samples.

This finalized non-member dataset serves as the foundation for feature extraction and attack model training in the subsequent stages.

3.3. Multi-shadow model fusion and feature enhancement mechanism

In black-box federated learning attacks, constructing effective attack features is particularly challenging due to the unknown architecture of the target model and the limited accessible output information. Existing methods typically rely on the softmax outputs of a single shadow model or their derived low-dimensional statistical features, which struggle to effectively mitigate generalization errors caused by model heterogeneity and predictive uncertainty. This limitation ultimately restricts the upper bound of attack performance.

To overcome the above challenges, we propose a feature enhancement mechanism based on collaborative output fusion from multiple shadow models. Specifically, we construct a set of structurally consistent yet independently trained shadow models and apply a high-dimensional softmax concatenation strategy to jointly model the behavioral differences across model responses. This method leverages a model ensemble perspective to mine multi-source predictive information, significantly improving the stability, discriminability, and robustness of the attack features. As a result, it provides more expressive input representations for membership inference attacks in black-box settings.

3.3.1. Multi-shadow cooperative modeling framework

To effectively simulate the prediction behaviors of a target model with unknown architecture and data distribution, we propose a Multi-Shadow Cooperative Modeling Framework. This framework constructs multiple independently trained model subsets on an auxiliary dataset to approximate the behavioral patterns and probabilistic outputs of the black-box target model.

Let the auxiliary dataset D_{aux} be partitioned into N non-overlapping subsets $\{D_1, D_2, \dots, D_N\}$, where each subset D_i is used to independently train a shadow model M_i . These models form the shadow model set $S = \{M_1, M_2, \dots, M_N\}$. All models adopt an identical architecture but are trained independently, thereby ensuring behavioral diversity while maintaining structural comparability:

$$M_i : \mathbb{R}^d \rightarrow \mathbb{R}^C, \quad i = 1, \dots, N \quad (8)$$

Here, d denotes the dimensionality of the input samples, and C is the number of classification categories.

Perspective of collaborative modeling across multiple models. In standard machine learning, the prediction of a single model for an input sample x typically reflects its empirical inductive inference based on a specific training distribution. To mitigate local data bias and capture richer discriminative boundary information, we adopt a collaborative modeling approach across multiple models and introduce the notion of a “multi-model perturbation space” defined as:

$$P(x) = \{p_i(x)\}_{i=1}^N, \quad p_i(x) = \text{softmax}(M_i(x)) \in \mathbb{R}^C \quad (9)$$

The set $P(x)$ can be interpreted as a family of probabilistic responses of the sample x under N neighborhood distributions induced by the shadow models. Together, they form a “Model Perturbation Response Track (MPRT)” in the C -dimensional probability space, which reveals the sample’s uncertainty and separability along the predictive landscape.

Statistical coverage and behavioral discrepancy of response distributions. The core advantage of the collaborative training structure lies in the output discrepancy among models and the broad statistical coverage of training distributions. According to ensemble learning theory, if the shadow models satisfy the following assumptions:

- **Output Heterogeneity Assumption:** $\exists i, j$ s.t. $\mathbb{E}_{x \in \mathcal{X}} [\|p_i(x) - p_j(x)\|_2] > \delta$
- **Coverage Compensation Assumption:** $\cup_i D_i \approx D_{\text{target}}$

then the model set S can construct an approximated ensemble structure $\hat{M}_{\text{approx}}$ in the response space that approximates the behavior of the target model, thereby enhancing the stability and generalizability of the attack model training.

Definition of the representation space of multi-model perturbation paths. Furthermore, the softmax response of each shadow model on a sample x can be regarded as a point in the probability space Δ^C , defined as:

$$p_i(x) \in \Delta^C = \left\{ \mathbf{p} \in \mathbb{R}^C \mid \sum_{j=1}^C p_j = 1, p_j \geq 0 \right\} \quad (10)$$

The response sequence of multiple shadow models $\{p_i(x)\}_{i=1}^N$ constitutes a dynamic perturbation path:

$$\gamma(x) : \{1, \dots, N\} \rightarrow \Delta^C, \quad \gamma(i) = p_i(x) \quad (11)$$

This path describes the probabilistic behavior trajectory of sample x across multiple models. Its statistical characteristics — such as entropy, mean, and standard deviation — can serve as the foundation for subsequent attack feature extraction (see Section 3.3.3).

3.3.2. Concatenated softmax fusion mechanism

In constructing features for membership inference attacks, a critical challenge lies in systematically preserving fine-grained behavioral information embedded in model responses. Traditional approaches, such as single-model feature extraction or mean-based fusion, inherently rely on low-dimensional compression mappings, which limit the ability to exploit output heterogeneity across models.

To address this limitation, we introduce a structurally expressive and semantically rich mechanism termed **Concatenated Softmax Fusion (CSF)**. This mechanism enables a comprehensive modeling of response behaviors through formalizations rooted in vector space composition, tensor algebra, and nonlinear mapping theories.

1. Vector concatenation and space construction. Given a set of shadow models $\mathcal{M} = \{M_i\}_{i=1}^N$, where each model produces a softmax output vector $p_i(x) \in \mathbb{R}^C$, the concatenation operation is defined as:

$$P(x) = \bigoplus_{i=1}^N p_i(x) = \begin{bmatrix} p_1(x) \\ p_2(x) \\ \vdots \\ p_N(x) \end{bmatrix} \in \mathbb{R}^{C \cdot N} \quad (12)$$

The concatenated vector $P(x)$ realizes a full-scale aggregation of model responses. Essentially, it embeds multiple independent softmax outputs into a unified joint vector space:

$$P(x) \in S_N := \bigotimes_{i=1}^N \mathbb{R}_i^C \quad (13)$$

Here, S_N denotes the tensor product subspace of N category probability spaces. This structure can be viewed as the direct product space $(\mathbb{R}^C)^N$, with a dimensionality of $C \cdot N$, which retains more degrees of freedom in the output responses than any form of weighted fusion mapping.

2. Algebraic expansion and index mapping of concatenated vectors. Let the concatenated output be expressed as:

$$P(x) = [p_1^{(1)}(x), \dots, p_1^{(C)}(x), p_2^{(1)}(x), \dots, p_N^{(C)}(x)]^\top \quad (14)$$

where $p_i^{(j)}(x)$ denotes the softmax prediction value of model M_i for class j . We define an index mapping function:

$$\Phi : \{1, \dots, N\} \times \{1, \dots, C\} \rightarrow \{1, \dots, C \cdot N\}, \quad \Phi(i, j) = (i-1) \cdot C + j \quad (15)$$

This indexing mechanism guarantees that the concatenated structure can be unfolded into a fixed-dimensional tensor, facilitating vectorization and subsequent discriminative function modeling.

3. Embedding in feature space and discriminator definition. The concatenation mapping is defined as:

$$F : \mathcal{X} \rightarrow \mathbb{R}^{C \cdot N}, \quad x \mapsto P(x) \quad (16)$$

This embedding operation maps the original sample space $\mathcal{X}$ into a $C \cdot N$ -dimensional feature space, within which a membership inference discriminator f can be defined:

$$f(x) = h(P(x); \theta), \quad h : \mathbb{R}^{C \cdot N} \rightarrow \{0, 1\} \quad (17)$$

where θ denotes the parameters of the attack model. Common choices for h include nonlinear classification functions based on tree-based models, support vector machines, or neural networks.

4. High-dimensional separability and generalization foundation. The concatenated structure constructs a comprehensive space across multi-model responses, enabling the attack model to observe prediction consistency and distributional shift across models:

$$A_{\text{var}}(x) = \frac{1}{C} \sum_{j=1}^C \text{Var} \left(\{p_i^{(j)}(x)\}_{i=1}^N \right) \quad (18)$$

In addition, structural indicators such as output variance and average entropy can serve as auxiliary features:

$$H_{\text{avg}}(x) = \frac{1}{N} \sum_{i=1}^N \left(- \sum_{j=1}^C p_i^{(j)}(x) \log p_i^{(j)}(x) \right) \quad (19)$$

$$\sigma(x) = \text{Std}(P(x)) \quad (20)$$

These constructions provide statistical characterizations of model response fluctuation, serving as a structural foundation for improving the robustness of the attack discriminator.

3.3.3. Statistical attack feature extraction mechanism

Although the high-dimensional concatenated vector $P(x) \in \mathbb{R}^{C \cdot N}$ already contains complete model response information, directly using it for training may lead to the curse of dimensionality, overfitting, and susceptibility to noise. Therefore, this work further introduces a **Statistical Feature Projection (SFP)** [28] mechanism based on the high-dimensional concatenation, employing nonlinear feature mappings to extract discriminative mid-dimensional attack feature representations.

1. Feature mapping definition and space compression. Let $P(x) = [p_1(x) \parallel p_2(x) \parallel \dots \parallel p_N(x)] \in \mathbb{R}^{C \cdot N}$ denote the concatenated vector. We define a set of statistical feature extraction functions $\mathcal{T} = \{\phi_k\}_{k=1}^d$, where each $\phi_k : \mathbb{R}^{C \cdot N} \rightarrow \mathbb{R}$ represents a statistical mapping function.

The final input feature vector to the attack model is then:

$$F(x) = [\phi_1(P(x)), \phi_2(P(x)), \dots, \phi_d(P(x))]^\top \in \mathbb{R}^d \quad (21)$$

where $d \ll C \cdot N$, forming a compact yet highly discriminative feature representation space.

2. Construction of core statistical features. This work extracts seven types of statistical features from the softmax outputs of each shadow model as the basis of the attack feature vector, including maximum confidence, confidence margin, entropy, standard deviation, second-highest confidence, confidence mean, and the ratio of maximum confidence to mean. These are designed to characterize model confidence, prediction uncertainty, distribution smoothness, and inter-class margin diversity, providing a comprehensive perspective of the model's behavioral responses.

3. Input format of the attack model training data. The final training data for the attack model is constructed as:

$$\mathcal{D}_{\text{attack}} = \{(F(x_i), y_i)\}_{i=1}^{N_{\text{train}}} \quad (22)$$

where $F(x_i)$ represents the attack feature vector extracted from the multi-shadow softmax outputs for the i th sample, and $y_i \in \{0, 1\}$ is the corresponding membership label.

Overall, the extracted statistical features provide a compact yet expressive representation, laying the foundation for subsequent attack model optimization.

3.4. Attack model construction and optimization mechanism

After constructing high-dimensional attack features as described in the previous sections, the core task is transformed into building a discriminative function $A : \mathbb{R}^d \rightarrow \{0, 1\}$ to accurately classify member and non-member samples. Considering the challenges of noisy training data, class imbalance, and high-dimensional input features, the design of the attack model must balance model complexity, generalization ability, and interpretability. In this work, we adopt the **Extreme Gradient Boosting (XGBoost)** framework to construct the attack model and introduce a series of optimization mechanisms to maximize its discriminative power and robustness.

3.4.1. Model selection and structural definition

The gradient boosting decision tree (GBDT) model can be regarded as a nonlinear extension of additive models, whose output is a linear combination of multiple base regression trees:

$$A(x) = \sigma\left(\sum_{t=1}^T f_t(x)\right), \quad f_t \in \mathcal{F} \quad (23)$$

where $\sigma(\cdot)$ denotes the sigmoid activation function, $\mathcal{F}$ represents the space of regression trees, and T is the number of boosting rounds. During training, the model minimizes the residual error at each iteration, by applying a second-order Taylor expansion to approximate the loss function:

$$\mathcal{L}^{(t)} = \sum_{i=1}^n \left[g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i) \right] + \Omega(f_t) \quad (24)$$

Here: $g_i = \frac{\partial \ell(y_i, A_{t-1}(x_i))}{\partial A_{t-1}(x_i)}$, $h_i = \frac{\partial^2 \ell(y_i, A_{t-1}(x_i))}{\partial A_{t-1}^2(x_i)}$

represent the first and second derivatives of the loss function with respect to the model output. The regularization term $\Omega(f_t)$ controls the complexity of the tree structure.

The decision node split is determined by maximizing the split gain:

$$\text{Gain} = \frac{1}{2} \left(\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right) - \gamma \quad (25)$$

This mechanism guarantees that splits are performed only when it yields significant information gain, thereby enhancing the robustness of the structural optimization.

3.4.2. Imbalanced sample handling and distribution augmentation

The attack modeling process suffers from a severe class imbalance problem, where the number of member samples significantly exceeds that of non-member samples. Let the positive class denote member samples and the negative class denote non-member samples. The ratio between positive and negative samples is given by:

$$r = \frac{|\mathcal{D}_1|}{|\mathcal{D}_0|} \ll 1 \quad (26)$$

Such imbalance leads to biased learning toward the member class and degrades the attack model's discriminative capability and generalization. To alleviate this issue, we introduce an improved version of the SMOTE algorithm (Synthetic Minority Over-sampling Technique). SMOTE generates synthetic minority samples by linearly interpolating between neighboring instances in the feature space, thereby expanding the data manifold of the minority class without simply duplicating data. This interpolation-based augmentation preserves local geometric relationships and avoids overfitting caused by naive resampling. In our framework, SMOTE is applied to non-member features to enhance their distributional diversity and to balance the class proportions during the attack model training. Formally, each new sample is generated as:

$$x_{\text{new}}^{(i)} = x^{(i)} + \lambda \cdot (x_{\text{nn}}^{(i)} - x^{(i)}), \quad \lambda \sim \mathcal{U}(0, 1) \quad (27)$$

After performing K iterations, the augmented non-member sample set is constructed as:

$$\tilde{\mathcal{D}}_0 = \mathcal{D}_0 \cup \bigcup_{k=1}^K \text{SMOTE}^{(k)}(\mathcal{D}_0) \quad (28)$$

According to the theory of generalization error decomposition:

$$\mathcal{E}_{\text{gen}} = \mathcal{E}_{\text{approx}} + \mathcal{E}_{\text{estimation}} + \mathcal{E}_{\text{optimization}} \quad (29)$$

By enlarging the non-member sample space, the proposed SMOTE-based augmentation effectively reduces the estimation error term $\mathcal{E}_{\text{estimation}}$ and improves the generalization ability of the attack model. Compared with random duplication or noise-based oversampling, SMOTE explicitly models the feature-space continuity of non-member data, enabling smoother and more stable decision boundaries between member and non-member classes.

3.4.3. Generalization evaluation and feature sensitivity analysis

The effectiveness of the attack model ultimately depends on a systematic evaluation of its generalization performance. To comprehensively validate the stability and practicality of the proposed framework under high-dimensional input and weak supervision scenarios, we conduct in-depth analysis from multiple perspectives on both the generalization capability and feature sensitivity of the attack model. Specifically, we adopt the following three core metrics:

- **Accuracy:** Measures the overall classification capability of the attack model in distinguishing member and non-member samples.

$$\text{Acc} = \frac{TP + TN}{TP + TN + FP + FN} \quad (30)$$

- **AUC (Area Under the ROC Curve):** Evaluates the model's ability to rank samples. A higher AUC indicates better discrimination between positive and negative samples.
- **TPR@FPR:** The true positive rate under a fixed false positive rate α , commonly used to assess performance in low-FPR scenarios:

$$\text{TPR}_\alpha = \frac{TP}{TP + FN} \Big|_{\text{FPR} \leq \alpha} \quad (31)$$

To construct input features for the attack model, we extract a $7 \times S$ -dimensional statistical feature vector from the softmax outputs of S shadow models. These features include: maximum confidence, second-highest confidence, confidence gap, mean confidence, standard deviation, confidence ratio, and entropy. Such statistics are capable of capturing distributional differences between member and non-member samples.

Although some studies introduce complex feature selection techniques (e.g., mutual information filtering, dimensionality reduction projections), we deliberately avoid such explicit compression mechanisms for the following reasons:

- The seven proposed statistical indicators already capture discriminative properties across various dimensions of the output distribution;
- High-dimensional concatenation of multiple models' outputs enhances the robustness against feature redundancy and reduces reliance on any specific model;
- Introducing artificial dimensionality reduction or weighting mechanisms may bring additional hyperparameter tuning burdens and harm feature generality.

Therefore, we retain the complete set of statistical features from all shadow models as input to the attacker and further apply SMOTE-based augmentation to enhance model generalization. Experimental results on the validation set demonstrate stable and superior attack performance compared to baseline methods, indicating that the constructed high-dimensional feature representations exhibit strong discriminative power and generalization ability.

4. Experimental design and result analysis

4.1. Experimental settings

This section introduces the overall experimental environment configuration and dataset partitioning strategy used in this study.

4.1.1. Experimental environment

All experiments were conducted on a server equipped with an NVIDIA RTX 4080 GPU, running Ubuntu 22.04. The deep learning framework used was PyTorch 2.0 with Python 3.10. Common libraries employed in the experiments include torchvision, xgboost, imbalanced-learn, and scikit-learn. To ensure reproducibility of results, a fixed random seed was set during both training and evaluation phases.

4.1.2. Dataset partitioning strategy

We conduct membership inference attacks on four commonly used image classification datasets: CIFAR-10, CIFAR-100, Fashion-MNIST, and SVHN. To ensure consistency and comparability across experiments, a unified partitioning strategy is applied.

For each dataset, the first 30,000 images from the training set are selected as member samples, which are used to participate in federated

training and serve as the positive class (member) features for the attack model. The remaining training samples, together with the official test set, form the basic non-member sample set. In addition, pseudo non-member samples are generated using DCGAN and filtered based on the discriminator score (retaining those with a score below 0.6) to enhance the diversity and generalization of the non-member set. The final non-member class for attack modeling consists of these three components and is treated as the negative class (non-member).

After feature extraction for both member and non-member samples, we apply the SMOTE algorithm to address class imbalance by over-sampling, thereby constructing a balanced training set for the attack model. This partitioning strategy is kept consistent across all datasets to ensure fair comparison and performance evaluation of different attack methods across various data domains.

4.2. Experimental methodology

To systematically evaluate the performance of the proposed attack framework, we design a series of experiments on four datasets — CIFAR-10, CIFAR-100, Fashion-MNIST, and SVHN — including target attack validation, generalization analysis, and ablation studies. The overall experimental methodology is described as follows:

4.2.1. Target attack design

We construct federated learning target models on CIFAR-10, CIFAR-100, Fashion-MNIST, and SVHN datasets using ResNet-18 [29] as the backbone architecture. Attack experiments are conducted under three representative federated learning frameworks: FedAvg [30], FedMD [31], and FedProx [32]. The federated training hyperparameters are configured as follows:

- Number of clients: 5;
- Local training epochs per round: 5;
- Local batch size: 64;
- Global communication rounds: set to 30 and 50 respectively;
- Optimizer: Adam, with an initial learning rate of 0.001.

4.2.2. Baseline method migration and reproduction

For fair comparison, three representative black-box membership inference attack methods are adapted to our unified experimental environment for reproduction and evaluation, including:

- **Shokri Attack:** The first work that introduced membership inference attacks. It constructs an attack classifier based on multiple shadow models, using softmax outputs as features to determine sample membership status.
- **He et al. (2024) Attack:** This method performs membership inference by modeling the temporal pattern of the maximum confidence scores over time, using confidence sequences to train a sequence classifier.
- **MIB Attack:** A membership inference attack based on poisoning-trigger mechanisms. It implants triggers into member samples and detects the target model's response to these triggers for membership prediction.

4.3. Experimental settings

4.3.1. Attack performance under the FedAvg protocol

To evaluate the adaptability and effectiveness of the proposed attack method under a typical federated learning framework, we focus on experiments conducted using the widely adopted FedAvg protocol. Owing to its efficiency and ease of deployment, FedAvg is commonly used in homogeneous model settings. However, its periodic parameter aggregation mechanism may introduce potential privacy leakage risks. Experiments are conducted on CIFAR-10, CIFAR-100, Fashion-MNIST, and SVHN datasets under four configurations combining different numbers of communication rounds (30, 50) and shadow models

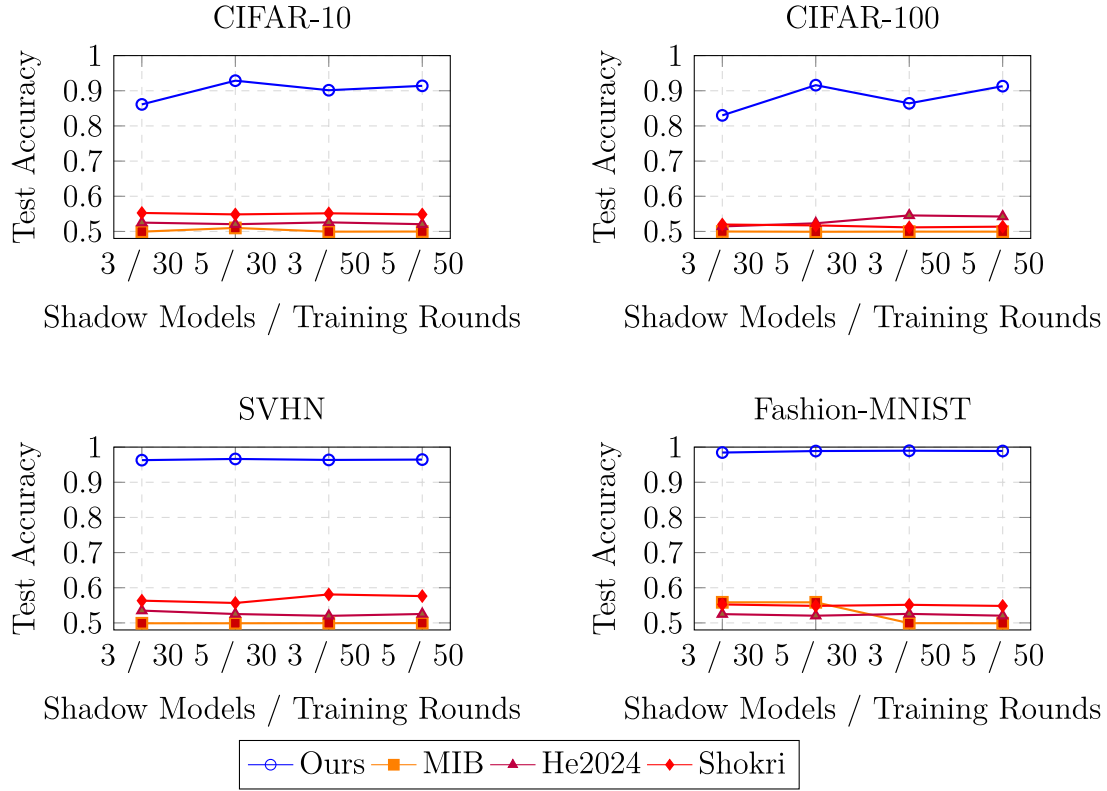


Fig. 4. Test accuracy comparison on four datasets (Ours vs. MIB, He2024, Shokri).

(3, 5). We systematically compare the proposed method with three representative black-box attacks (He2024, MIB, Shokri) in terms of Accuracy, TPR@FPR=1%, and AUC. Table 3 (see Appendix A) summarizes the comprehensive performance comparison under the FedAvg protocol, including Accuracy, TPR@FPR=1%, and AUC across different datasets and attack configurations. To maintain brevity in the main text, detailed numerical results are provided in the appendix.

As shown in Table 3 and Fig. 4, the proposed method significantly outperforms existing baseline attacks under the FedAvg protocol, achieving the highest Accuracy and AUC across all datasets and configurations. For example, on the CIFAR-10 dataset, our method achieves an attack accuracy of 0.9140 and an AUC of 0.9931 under the configuration of 5 shadow models and 50 communication rounds, which clearly exceeds that of He2024 (0.5205) and MIB (0.4996). On the SVHN dataset, the accuracy of our method consistently remains above 0.96, with the AUC approaching 0.99, whereas traditional methods generally fall below 0.55. Similar trends are observed on Fashion-MNIST and CIFAR-100, indicating that the proposed method is not only well-suited for high-dimensional color image tasks (e.g., CIFAR-100) but also maintains superior performance on grayscale, low-resolution tasks.

Moreover, the results reveal two additional insights: (1) Increasing the number of communication rounds from 30 to 50 significantly enhances the attack performance of our method, suggesting that increased communication allows deeper memorization of member patterns, hence boosting attack success; and (2) Increasing the number of shadow models from 3 to 5 also improves performance, validating the effectiveness of the high-dimensional concatenation fusion mechanism in capturing member behavior. In contrast, the Shokri method shows limited sensitivity to both communication rounds and shadow model scale, reflecting its constrained feature representation capability.

4.3.2. Attack performance under the FedMD protocol

To evaluate the generalizability and robustness of the proposed method across different federated learning protocols, we reproduce

the complete attack pipeline under the FedMD framework. As a representative knowledge distillation-based protocol, FedMD aggregates knowledge via shared unlabeled data, which enhances privacy protection and supports model heterogeneity, while also increasing the difficulty of membership inference attacks. The experimental settings are kept consistent with those under the FedAvg protocol, covering four image classification datasets and using configurations that vary the number of communication rounds (30, 50) and shadow models (3, 5). We compare the performance of our method with three baseline approaches: He2024, MIB, and Shokri. The complete comparison results are provided in Appendix Table 4, covering key metrics such as attack accuracy, AUC, and TPR@FPR=1%. Fig. 5 illustrates the accuracy trends of different methods under various configurations, further highlighting the stability and superiority of the proposed method.

As shown in Table 4 and Fig. 5, despite the challenges introduced by model heterogeneity and collaborative training via knowledge distillation, the proposed attack method consistently maintains a leading advantage across all datasets, significantly outperforming the three representative baselines: He2024, MIB, and Shokri.

Specifically, on CIFAR-10 and CIFAR-100, our method achieves over 0.85 in attack accuracy and above 0.95 in AUC under most configurations, demonstrating strong attack capability for high-dimensional color image tasks. For relatively simpler grayscale tasks such as SVHN and Fashion-MNIST, the method consistently achieves an accuracy between 0.96 and 0.98, with AUC scores nearing 0.99, substantially outperforming the MIB and Shokri baselines, which generally remain within the 0.50–0.55 range.

Notably, although the He2024 method shows a slight advantage in certain settings (e.g., achieving a higher TPR@FPR=1%), its overall accuracy and generalization performance are considerably inferior to our method, indicating potential overfitting or instability in its attack model.

Moreover, the experiments reveal that increasing the number of communication rounds from 30 to 50 or the number of shadow models

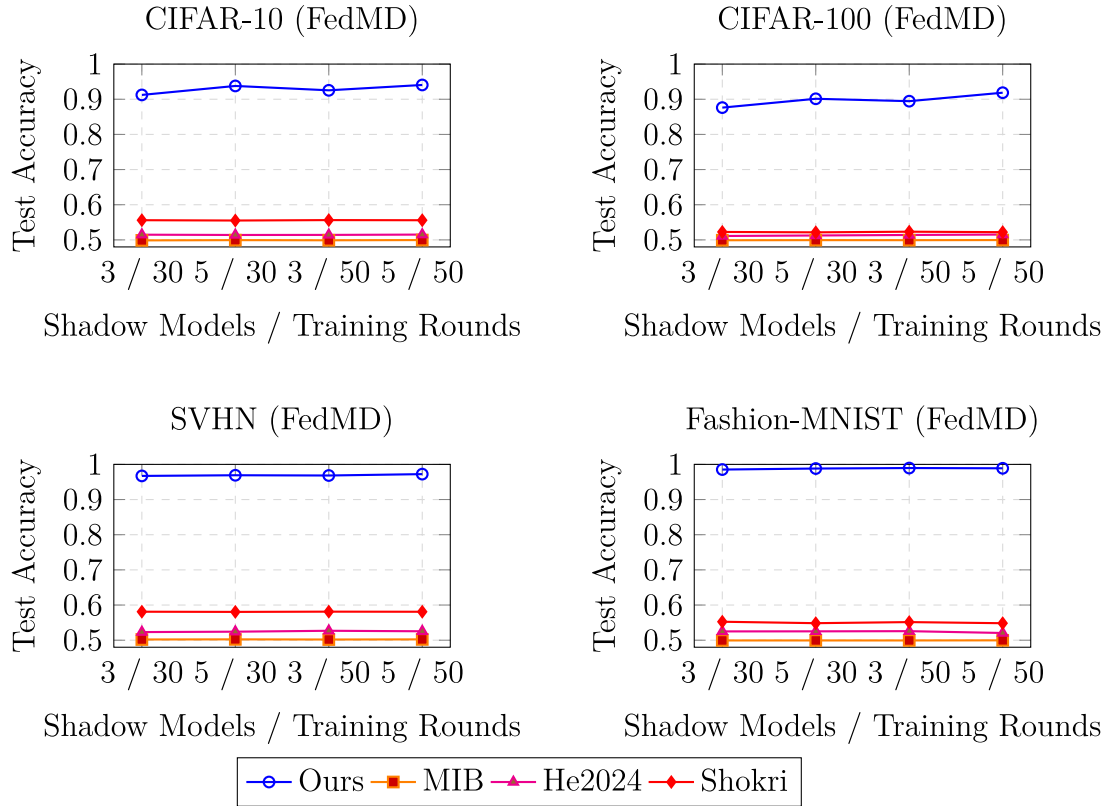


Fig. 5. Test accuracy comparison on four datasets under FedMD protocol.

from 3 to 5 consistently improves the performance of our method across all datasets, demonstrating that the proposed feature construction mechanism generalizes well even in knowledge distillation scenarios. In contrast, traditional methods are more sensitive to model architecture and distillation variations, exhibiting greater performance fluctuations.

4.3.3. Attack performance under the FedProx protocol

To evaluate the adaptability of the proposed method under the FedProx protocol, we reproduce the complete attack pipeline using the same experimental configurations as in the FedAvg and FedMD settings. FedProx introduces an ℓ_2 regularization term during local training to constrain deviation from the global model, aiming to mitigate system heterogeneity caused by non-independent and identically distributed (non-IID) data and to enhance convergence stability. However, such consistency may diminish the behavioral divergence between member and non-member samples, thus increasing the complexity of effective membership inference. Experiments are conducted on CIFAR-10, CIFAR-100, Fashion-MNIST, and SVHN across four configurations defined by shadow model numbers $N = 3, 5$ and communication rounds of 30 and 50. We systematically compare the performance of our method with baseline approaches including He2024, MIB, and Shokri. The complete results are presented in Appendix Table 5, covering key metrics such as Accuracy, AUC, and TPR@FPR=1%. Fig. 6 further illustrates the trends in attack accuracy under different configurations, demonstrating the stability and generalizability of the proposed method in heterogeneous environments.

As shown in Table 5 and Fig. 6, the proposed attack method consistently outperforms representative baselines such as He2024, MIB, and Shokri under the FedProx protocol, achieving the highest attack accuracy and AUC across all configurations on the four standard image classification datasets. For example, on the CIFAR-100 dataset, when

using 5 shadow models and 50 communication rounds, our method achieves an accuracy of 0.9113 and an AUC of 0.9627, significantly exceeding He2024 (0.5525) and MIB (0.4992). On the SVHN dataset, the proposed method maintains an accuracy above 0.96 and an AUC close to 0.99 under all settings, whereas the performance of traditional methods such as Shokri and MIB fluctuates around random levels (approximately 0.50). Similar trends are observed on Fashion-MNIST and CIFAR-10, indicating that our method generalizes well to both low-dimensional grayscale images and complex color image tasks.

Moreover, the experiments reveal two key trends: First, increasing the number of communication rounds from 30 to 50 significantly improves attack performance, suggesting that increased training interactions accentuate membership-specific behaviors. Second, increasing the number of shadow models from 3 to 5 further enhances the performance, validating the effectiveness of the multi-model fusion mechanism. In contrast, the Shokri method shows limited sensitivity to these variations, indicating its constrained attack capability.

Overall, the consistent superiority of our framework across FedAvg, FedMD, and FedProx settings can be attributed to the following synergistic mechanisms. These results collectively demonstrate that the effectiveness of our proposed framework arises from the synergistic interaction of three key mechanisms: the discriminator-guided filtering ensures the authenticity and diversity of pseudo non-member samples, reducing label noise and stabilizing decision boundaries; the multi-shadow-model fusion enriches the attack feature space by integrating diverse prediction behaviors, thereby capturing more nuanced membership cues; and the SMOTE-based balancing strategy mitigates class imbalance and enhances generalization, providing a causal explanation for the observed performance improvements across datasets and federated protocols.

4.3.4. Generalization analysis of the attack model

Given that the structure of the attacker remains consistent across different protocols, and considering that FedAvg is the most classical

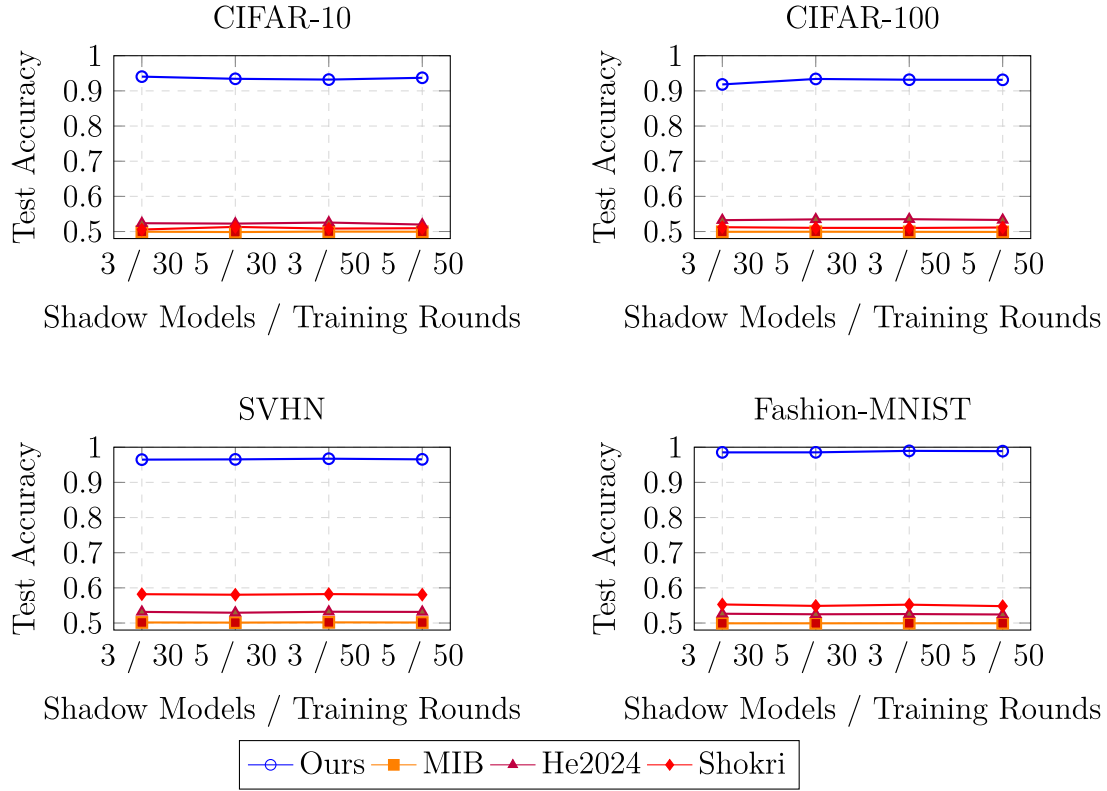


Fig. 6. Test accuracy comparison on four datasets under FedProx protocol.

Table 1

Validation of attack model generalization ability.

Dataset	Client/Shadow/Rounds	Train accuracy	Validation accuracy
CIFAR-10	5 clients/3 shadow/30 rounds	0.9882	0.9828
CIFAR-10	5 clients/5 shadow/30 rounds	0.9410	0.9162
CIFAR-10	5 clients/3 shadow/50 rounds	0.9647	0.9526
CIFAR-10	5 clients/5 shadow/50 rounds	0.9307	0.9106
CIFAR-100	5 clients/3 shadow/30 rounds	0.9158	0.8901
CIFAR-100	5 clients/5 shadow/30 rounds	0.9465	0.9305
CIFAR-100	5 clients/3 shadow/50 rounds	0.8953	0.8721
CIFAR-100	5 clients/5 shadow/50 rounds	0.9299	0.9052

and representative training protocol in federated learning, we evaluate the generalization ability of the attack model under the FedAvg setting by comparing its performance on the training and validation sets. This evaluation also indirectly reflects the robustness of the attacker under other federated protocols. The results are summarized in Table 1, and the corresponding accuracy comparison is illustrated in Fig. 7.

The results indicate that the accuracy gap between the training and validation sets is generally small (less than 3%) and both remain at a high level, suggesting that the attack model possesses good generalization ability without significant overfitting issues.

4.3.5. Ablation study analysis

To further investigate the contribution of each key module to the overall performance, we conduct a series of ablation experiments on the CIFAR-10 dataset. Table 2 presents the attack performance under different ablation settings, and the corresponding performance comparison is visualized in the line chart shown in Fig. 8.

As shown in Table 2 and Fig. 8, the following observations can be made:

- **The pseudo non-member sample generation and filtering mechanism** — including GAN-based synthesis and discriminator-based quality control — plays a pivotal role in determining attack efficacy. Removing these components leads to a significant drop in both accuracy and TPR@FPR=1%, highlighting the essential role of high-quality negatives in label integrity and attack generalization.
- **The number of shadow models and the fusion strategy** significantly affect the attack results. A single shadow model captures only one perspective of model behavior, whereas combining multiple shadow models enables richer representations from different angles, demonstrating that multi-model fusion effectively improves the discriminative power of attack features.

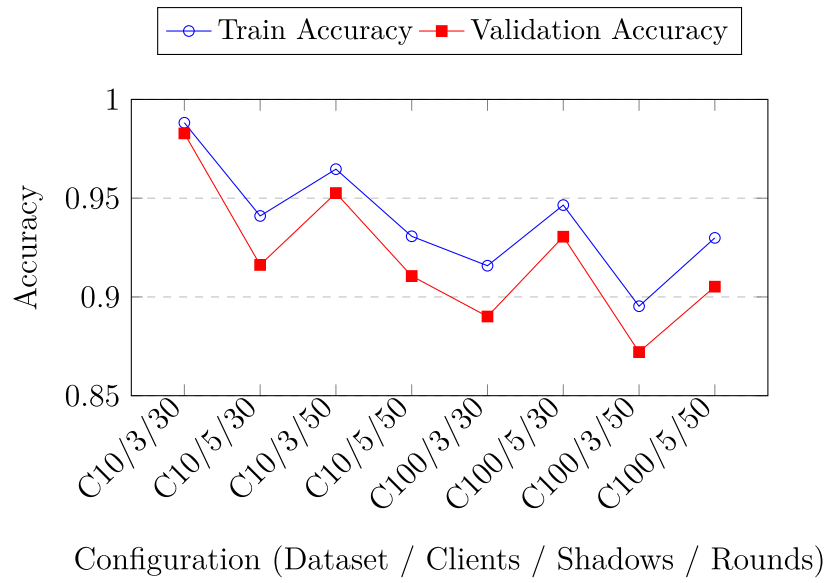


Fig. 7. Comparison of Training and Validation Accuracy for Generalization Verification.

Table 2

Ablation study results (CIFAR-10, 5 clients, 3 shadow models, 30 rounds).

Method	Test accuracy	TPR@FPR=1%	AUC
Full Method	0.8612	0.7581	0.9423
Remove GAN Samples	0.5099	0.0111	0.5119
Remove Discriminator Filtering	0.5708	0.0131	0.5356
Single Shadow + Statistical Features	0.8054	0.2314	0.8281
Single Shadow + Max Confidence	0.7471	0.1218	0.7413
Multi-Shadow + Concatenated Max Confidence	0.7844	0.1587	0.7863

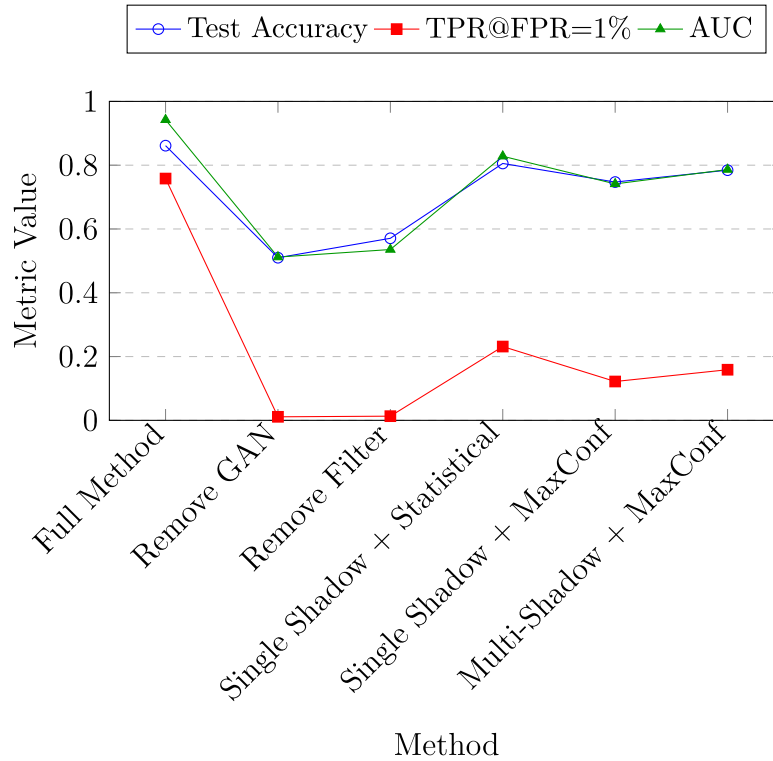


Fig. 8. Performance Comparison under Ablation Settings.

- **The structure of attack feature representation** also plays an important role in performance. When using only the maximum confidence (Max Confidence) as the attack vector — regardless of single or multiple models — the performance is notably inferior compared to using the full 7-dimensional statistical feature vector, confirming the importance of high-dimensional joint statistical representation in enhancing attack discriminability.

In summary, the proposed attack framework forms a complementary enhancement mechanism through pseudo non-member generation and filtering, multi-shadow model fusion, and statistical feature construction with concatenated representation. This significantly improves the robustness and accuracy of the attack model, demonstrating the synergistic efficacy of the overall attack design in balancing robustness, accuracy, and generalization.

4.4. Experimental summary

In summary, we conducted a comprehensive set of experiments on four representative image classification datasets — CIFAR-10, CIFAR-100, Fashion-MNIST, and SVHN — under three mainstream federated learning protocols: FedAvg, FedMD, and FedProx. The experiments include target attack performance comparisons, generalization ability evaluation, and ablation studies, which collectively validate the effectiveness and generality of the proposed black-box MIA enhancement framework based on pseudo non-member filtering and multi-shadow model softmax output fusion. The main experimental findings are as follows:

- **Superior attack performance across multiple protocols and datasets:** Compared to existing representative black-box attack methods (e.g., MIB and He et al.), the proposed method achieves significant improvements in key metrics such as Accuracy, TPR@FPR=1%, and AUC. On the CIFAR-10 dataset, it achieves up to 0.9812 in accuracy and 0.9767 in AUC. It also demonstrates stable performance on complex tasks such as CIFAR-100 and non-natural image datasets such as SVHN, showing strong attack discriminability and robustness across protocols.
- **Joint improvement through high-dimensional fusion and statistical feature modeling:** By aggregating softmax outputs from multiple shadow models into a high-dimensional joint response space, and incorporating various statistical features (e.g., confidence entropy, standard deviation, and max-second-max gap), the proposed method effectively captures inter-model variations and enhances decision boundary complexity. This leads to significant performance gains over single-model or low-dimensional feature schemes.
- **Discriminator-based filtering effectively controls pseudo non-member quality:** By assigning discriminator scores to generated samples and applying a threshold (e.g., $\tau = 0.6$), overly “realistic” pseudo non-member samples are filtered out. This mitigates label pollution and distribution confusion, thereby improving the purity of non-member features and the stability of attack model training.
- **The attack model demonstrates strong generalization ability:** A dual-track evaluation mechanism based on training and validation sets is constructed, and combined with SMOTE-based augmentation to mitigate overfitting risks. Experiments show that the accuracy gap between training and validation sets is less than 3%, indicating good generalization performance in high-dimensional and weakly supervised settings.
- **Each module contributes clearly, and the design is systematically coherent and theoretically well-founded:** Ablation studies reveal that pseudo non-member generation, discriminator filtering, multi-shadow model fusion, and multi-dimensional feature construction all play significant roles in enhancing attack performance, validating the systematic design advantage across four dimensions: generation, filtering, fusion, and representation.

In conclusion, the proposed black-box MIA enhancement framework based on pseudo non-member filtering and multi-shadow model output fusion consistently achieves superior performance across diverse federated learning protocols and image classification tasks. It exhibits strong generalizability, structural scalability, and practical applicability, offering a unified and scalable solution for strengthening black-box inference capabilities in federated learning systems, with implications for both adversarial analysis and privacy auditing. Although the proposed framework demonstrates strong performance with high AUC and TPR@FPR=1% values, we also recognize several practical trade-offs in real-world deployment. First, the integration of multi-shadow-model fusion and discriminator-guided pseudo-sample filtering introduces additional computational overhead, particularly during feature extraction and GAN training stages. Second, while the cost remains acceptable in our current experimental scale, further optimization is required for large-scale federated systems with more participating clients, such as using lightweight generative architectures or parallelized training mechanisms. Third, the stability and quality of GAN-generated samples are sensitive to hardware resources, which may impact deployment in resource-constrained environments. Future research will explore adaptive thresholding and decentralized GAN coordination strategies to mitigate these limitations. Overall, these trade-offs reflect a balanced relationship between attack performance, computational complexity, and scalability, making the conclusions more practically relevant.

5. Conclusion and future work

We investigate membership inference attacks (MIA) in black-box federated learning and propose an enhanced framework to address challenges such as limited feature expressiveness, poor pseudo sample quality, and overfitting risk. The proposed method leverages pseudo non-member samples generated by a DCGAN and filtered by a discriminator to ensure data quality. To enrich attack features, softmax outputs from multiple shadow models are concatenated into high-dimensional vectors, from which statistical features are extracted. The final attacker is trained using SMOTE-based class balancing and optimized with XGBoost to enhance generalization and discriminative power. Extensive experiments on four benchmark datasets (CIFAR-10, CIFAR-100, Fashion-MNIST, SVHN) and three federated learning protocols (FedAvg, FedMD, FedProx) demonstrate that our method significantly outperforms state-of-the-art baselines (e.g., Shokri, He et al. and MIB) in terms of Accuracy, AUC, and TPR@FPR=1%, validating the effectiveness of multi-module integration in enhancing attack performance.

Despite its strong and stable performance across diverse scenarios, the proposed method still faces several limitations, including the instability of sample generation, the challenge of controlling the dimensionality of feature representation, and insufficient characterization of the attack generalization boundary. Future work may explore the following directions: (1) introducing more advanced generative models [33], such as diffusion models or variational autoencoders (VAE), to improve the diversity and separability of pseudo non-member samples; and (2) investigating the application of self-attention mechanisms [34] and Transformer architectures [35] in attack feature modeling.

Table 3

Performance comparison of membership inference attacks across CIFAR-10, CIFAR-100, Fashion-MNIST, and SVHN under FedAvg protocol.

Dataset	Shadows/Rounds	Method	Accuracy	TPR@FPR=1%	AUC
CIFAR-10	3/30	Ours	0.8612	0.7581	0.9423
CIFAR-10	3/50	Ours	0.9015	0.8149	0.9617
CIFAR-10	5/30	Ours	0.9287	0.8516	0.9728
CIFAR-10	5/50	Ours	0.9140	0.8692	0.9754
CIFAR-10	3/30	He2024	0.5250	0.8402	0.5260
CIFAR-10	3/50	He2024	0.5257	0.2428	0.5319
CIFAR-10	5/30	He2024	0.5252	0.8493	0.5271
CIFAR-10	5/50	He2024	0.5205	0.4400	0.5236
CIFAR-10	3/30	MIB	0.4993	0.0068	0.5279
CIFAR-10	3/50	MIB	0.4993	0.0057	0.5319
CIFAR-10	5/30	MIB	0.5006	0.0074	0.5234
CIFAR-10	5/50	MIB	0.4996	0.0059	0.5269
CIFAR-10	3/30	Shokri	0.5539	0.0137	0.5760
CIFAR-10	3/50	Shokri	0.5544	0.0138	0.5755
CIFAR-10	5/30	Shokri	0.5545	0.0133	0.5764
CIFAR-10	5/50	Shokri	0.5543	0.0145	0.5784
CIFAR-100	3/30	Ours	0.8300	0.7494	0.9267
CIFAR-100	3/50	Ours	0.8642	0.7840	0.9527
CIFAR-100	5/30	Ours	0.9159	0.8410	0.9531
CIFAR-100	5/50	Ours	0.9131	0.8684	0.9652
CIFAR-100	3/30	He2024	0.5142	0.5297	0.5209
CIFAR-100	3/50	He2024	0.5455	0.2245	0.5559
CIFAR-100	5/30	He2024	0.5231	0.2635	0.5289
CIFAR-100	5/50	He2024	0.5423	0.2918	0.5591
CIFAR-100	3/30	MIB	0.4997	0.0093	0.5187
CIFAR-100	3/50	MIB	0.4994	0.0091	0.5124
CIFAR-100	5/30	MIB	0.4990	0.0080	0.5187
CIFAR-100	5/50	MIB	0.4993	0.0086	0.5154
CIFAR-100	3/30	Shokri	0.5196	0.0121	0.5263
CIFAR-100	3/50	Shokri	0.5115	0.0125	0.5202
CIFAR-100	5/30	Shokri	0.5171	0.0121	0.5231
CIFAR-100	5/50	Shokri	0.5135	0.0113	0.5177
SVHN	3/30	Ours	0.9629	0.9714	0.9887
SVHN	3/50	Ours	0.9635	0.9703	0.9887
SVHN	5/30	Ours	0.9663	0.9727	0.9911
SVHN	5/50	Ours	0.9645	0.9704	0.9906
SVHN	3/30	He2024	0.5350	0.6701	0.5467
SVHN	3/50	He2024	0.5253	0.3254	0.5326
SVHN	5/30	He2024	0.5201	0.3456	0.5299
SVHN	5/50	He2024	0.5342	0.3432	0.5348
SVHN	3/30	MIB	0.5020	0.0071	0.5112
SVHN	3/50	MIB	0.5019	0.0074	0.5103
SVHN	5/30	MIB	0.5021	0.0069	0.5104
SVHN	5/50	MIB	0.5020	0.0072	0.5109
SVHN	3/30	Shokri	0.5801	0.0112	0.5011
SVHN	3/50	Shokri	0.5803	0.0109	0.5009
SVHN	5/30	Shokri	0.5810	0.0115	0.5016
SVHN	5/50	Shokri	0.5808	0.0117	0.5015
Fashion-MNIST	3/30	Ours	0.9845	0.9754	0.9839
Fashion-MNIST	3/50	Ours	0.9897	0.9799	0.9896
Fashion-MNIST	5/30	Ours	0.9888	0.9763	0.9899
Fashion-MNIST	5/50	Ours	0.9889	0.9799	0.9865
Fashion-MNIST	3/30	He2024	0.5250	0.8402	0.5260
Fashion-MNIST	3/50	He2024	0.5257	0.2428	0.5319
Fashion-MNIST	5/30	He2024	0.5252	0.8493	0.5271
Fashion-MNIST	5/50	He2024	0.5205	0.4400	0.5236
Fashion-MNIST	3/30	MIB	0.4993	0.0068	0.5279
Fashion-MNIST	3/50	MIB	0.4993	0.0057	0.5319
Fashion-MNIST	5/30	MIB	0.4993	0.0057	0.5319
Fashion-MNIST	5/50	MIB	0.4996	0.0059	0.5269
Fashion-MNIST	3/30	Shokri	0.5525	0.0115	0.5234
Fashion-MNIST	3/50	Shokri	0.5515	0.0146	0.5257
Fashion-MNIST	5/30	Shokri	0.5484	0.0112	0.5195
Fashion-MNIST	5/50	Shokri	0.5483	0.0123	0.5217

Table 4

Performance comparison of membership inference attacks across CIFAR-10, CIFAR-100, SVHN, and Fashion-MNIST under FedMD protocol.

Dataset	Shadows/Rounds	Method	Accuracy	TPR@FPR=1%	AUC
CIFAR-10	3/30	Ours	0.8863	0.8393	0.9655
CIFAR-10	3/50	Ours	0.9435	0.9237	0.9700
CIFAR-10	5/30	Ours	0.9370	0.9122	0.9773
CIFAR-10	5/50	Ours	0.9812	0.9766	0.9767
CIFAR-10	3/30	He2024	0.5328	0.0157	0.5470
CIFAR-10	3/50	He2024	0.5372	0.0572	0.5535
CIFAR-10	5/30	He2024	0.5308	0.0588	0.5419
CIFAR-10	5/50	He2024	0.5281	0.0163	0.5443
CIFAR-10	3/30	MIB	0.5002	0.0069	0.5438
CIFAR-10	3/50	MIB	0.4995	0.0061	0.5440
CIFAR-10	5/30	MIB	0.5007	0.0066	0.5436
CIFAR-10	5/50	MIB	0.5002	0.0064	0.5434
CIFAR-10	3/30	Shokri	0.5555	0.0126	0.5765
CIFAR-10	3/50	Shokri	0.5536	0.0132	0.5751
CIFAR-10	5/30	Shokri	0.5517	0.0124	0.5739
CIFAR-10	5/50	Shokri	0.5522	0.0128	0.5742
CIFAR-100	3/30	Ours	0.9217	0.8610	0.9594
CIFAR-100	3/50	Ours	0.9316	0.8921	0.9636
CIFAR-100	5/30	Ours	0.9362	0.9164	0.9740
CIFAR-100	5/50	Ours	0.9385	0.9282	0.9733
CIFAR-100	3/30	He2024	0.5367	0.0403	0.5473
CIFAR-100	3/50	He2024	0.5314	0.0392	0.5476
CIFAR-100	5/30	He2024	0.5306	0.0386	0.5464
CIFAR-100	5/50	He2024	0.5256	0.0391	0.5450
CIFAR-100	3/30	MIB	0.5001	0.0065	0.5447
CIFAR-100	3/50	MIB	0.4993	0.0064	0.5445
CIFAR-100	5/30	MIB	0.4992	0.0062	0.5443
CIFAR-100	5/50	MIB	0.4991	0.0063	0.5440
CIFAR-100	3/30	Shokri	0.5134	0.0126	0.5251
CIFAR-100	3/50	Shokri	0.5150	0.0131	0.5272
CIFAR-100	5/30	Shokri	0.5142	0.0127	0.5261
CIFAR-100	5/50	Shokri	0.5148	0.0129	0.5269
SVHN	3/30	Ours	0.9515	0.9680	0.9810
SVHN	3/50	Ours	0.9586	0.9719	0.9874
SVHN	5/30	Ours	0.9621	0.9754	0.9915
SVHN	5/50	Ours	0.9620	0.9751	0.9912
SVHN	3/30	He2024	0.5346	0.0514	0.5407
SVHN	3/50	He2024	0.5339	0.0508	0.5395
SVHN	5/30	He2024	0.5354	0.0521	0.5412
SVHN	5/50	He2024	0.5332	0.0503	0.5390
SVHN	3/30	MIB	0.5023	0.0072	0.5098
SVHN	3/50	MIB	0.5021	0.0073	0.5096
SVHN	5/30	MIB	0.5020	0.0070	0.5095
SVHN	5/50	MIB	0.5022	0.0074	0.5097
SVHN	3/30	Shokri	0.5811	0.0116	0.5012
SVHN	3/50	Shokri	0.5813	0.0117	0.5013
SVHN	5/30	Shokri	0.5812	0.0115	0.5011
SVHN	5/50	Shokri	0.5814	0.0118	0.5014
Fashion-MNIST	3/30	Ours	0.9873	0.9740	0.9874
Fashion-MNIST	3/50	Ours	0.9899	0.9794	0.9886
Fashion-MNIST	5/30	Ours	0.9906	0.9801	0.9901
Fashion-MNIST	5/50	Ours	0.9909	0.9805	0.9904
Fashion-MNIST	3/30	He2024	0.5256	0.0506	0.5311
Fashion-MNIST	3/50	He2024	0.5259	0.0498	0.5315
Fashion-MNIST	5/30	He2024	0.5261	0.0509	0.5317
Fashion-MNIST	5/50	He2024	0.5257	0.0497	0.5310
Fashion-MNIST	3/30	MIB	0.4992	0.0065	0.5236
Fashion-MNIST	3/50	MIB	0.4991	0.0062	0.5234
Fashion-MNIST	5/30	MIB	0.4993	0.0063	0.5238
Fashion-MNIST	5/50	MIB	0.4994	0.0061	0.5239
Fashion-MNIST	3/30	Shokri	0.5492	0.0118	0.5230
Fashion-MNIST	3/50	Shokri	0.5501	0.0119	0.5236
Fashion-MNIST	5/30	Shokri	0.5496	0.0120	0.5233
Fashion-MNIST	5/50	Shokri	0.5497	0.0117	0.5231

Table 5

Performance comparison of membership inference attacks across CIFAR-10, CIFAR-100, SVHN, and Fashion-MNIST under FedProx protocol.

Dataset	Shadows/Rounds	Method	Accuracy	TPR@FPR=1%	AUC
CIFAR-10	3/30	Ours	0.9405	0.9237	0.9700
CIFAR-10	3/50	Ours	0.9321	0.9122	0.9651
CIFAR-10	5/30	Ours	0.9343	0.9184	0.9682
CIFAR-10	5/50	Ours	0.9372	0.9205	0.9693
CIFAR-10	3/30	He2024	0.5236	0.0380	0.5275
CIFAR-10	3/50	He2024	0.5255	0.0175	0.5272
CIFAR-10	5/30	He2024	0.5225	0.0138	0.5272
CIFAR-10	5/50	He2024	0.5198	0.0243	0.5273
CIFAR-10	3/30	MIB	0.4995	0.0074	0.5279
CIFAR-10	3/50	MIB	0.4998	0.0070	0.5270
CIFAR-10	5/30	MIB	0.4985	0.0068	0.5269
CIFAR-10	5/50	MIB	0.4994	0.0072	0.5264
CIFAR-10	3/30	Shokri	0.5057	0.0109	0.5076
CIFAR-10	3/50	Shokri	0.5084	0.0123	0.5116
CIFAR-10	5/30	Shokri	0.5131	0.0116	0.5170
CIFAR-10	5/50	Shokri	0.5094	0.0101	0.5132
CIFAR-100	3/30	Ours	0.9211	0.8982	0.9512
CIFAR-100	3/50	Ours	0.9185	0.8921	0.9490
CIFAR-100	5/30	Ours	0.9156	0.8844	0.9455
CIFAR-100	5/50	Ours	0.9132	0.8797	0.9433
CIFAR-100	3/30	He2024	0.5175	0.0138	0.5303
CIFAR-100	3/50	He2024	0.5208	0.3057	0.5317
CIFAR-100	5/30	He2024	0.5230	0.1475	0.5312
CIFAR-100	5/50	He2024	0.5270	0.0545	0.5358
CIFAR-100	3/30	MIB	0.5000	0.0080	0.5106
CIFAR-100	3/50	MIB	0.5000	0.0084	0.5203
CIFAR-100	5/30	MIB	0.5000	0.0075	0.5140
CIFAR-100	5/50	MIB	0.4995	0.0078	0.5221
CIFAR-100	3/30	Shokri	0.5177	0.0133	0.5267
CIFAR-100	3/50	Shokri	0.5228	0.0145	0.5332
CIFAR-100	5/30	Shokri	0.5202	0.0143	0.5312
CIFAR-100	5/50	Shokri	0.5240	0.0135	0.5351
SVHN	3/30	Ours	0.9492	0.9299	0.9654
SVHN	3/50	Ours	0.9498	0.9378	0.9643
SVHN	5/30	Ours	0.9577	0.9313	0.9657
SVHN	5/50	Ours	0.9468	0.9435	0.9676
SVHN	3/30	He2024	0.6979	0.0467	0.5285
SVHN	3/50	He2024	0.6979	0.0220	0.5139
SVHN	5/30	He2024	0.6979	0.0735	0.5262
SVHN	5/50	He2024	0.6979	0.0182	0.5157
SVHN	3/30	MIB	0.6716	0.0104	0.5210
SVHN	3/50	MIB	0.6728	0.0104	0.5157
SVHN	5/30	MIB	0.6727	0.0087	0.5208
SVHN	5/50	MIB	0.6724	0.0111	0.5160
SVHN	3/30	Shokri	0.5051	0.0115	0.5084
SVHN	3/50	Shokri	0.5067	0.0102	0.5087
SVHN	5/30	Shokri	0.5076	0.0122	0.5119
SVHN	5/50	Shokri	0.5094	0.0114	0.5127
Fashion-MNIST	3/30	Ours	0.9602	0.8849	0.9674
Fashion-MNIST	3/50	Ours	0.9580	0.8791	0.9652
Fashion-MNIST	5/30	Ours	0.9591	0.8815	0.9660
Fashion-MNIST	5/50	Ours	0.9548	0.8773	0.9637
Fashion-MNIST	3/30	He2024	0.5714	0.2687	0.5214
Fashion-MNIST	3/50	He2024	0.5714	0.3388	0.5385
Fashion-MNIST	5/30	He2024	0.5704	0.0133	0.5316
Fashion-MNIST	5/50	He2024	0.5714	0.0128	0.5376
Fashion-MNIST	3/30	MIB	0.5714	0.0109	0.5109
Fashion-MNIST	3/50	MIB	0.5714	0.0123	0.5132
Fashion-MNIST	5/30	MIB	0.5714	0.0069	0.4904
Fashion-MNIST	5/50	MIB	0.5714	0.0079	0.5170
Fashion-MNIST	3/30	Shokri	0.5071	0.0118	0.5093
Fashion-MNIST	3/50	Shokri	0.5081	0.0108	0.5105
Fashion-MNIST	5/30	Shokri	0.5092	0.0110	0.5132
Fashion-MNIST	5/50	Shokri	0.5079	0.0125	0.5112

CRedit authorship contribution statement

Qiang Shi: Writing – review & editing, Validation, Supervision.
Luzhen Ren: Writing – original draft, Validation, Methodology, Investigation, Conceptualization.
Xinfeng He: Writing – review & editing, Supervision, Data curation.

Declaration of AI-assisted Technologies

The authors acknowledge the use of generative AI tools (e.g., ChatGPT by OpenAI) in assisting with English editing and grammar correction during the manuscript preparation. The authors reviewed and revised the content as needed and take full responsibility for the final version of the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Full performance comparison under FedAvg protocol

See Table 3.

Appendix B. Full performance comparison under FedMD protocol

See Table 4.

Appendix C. Full performance comparison under FedProx protocol

See Table 5.

Data availability

Data will be made available on request.

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